

Dr. Anubha Parashar

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Applied AI Engineer | LLM Agents | Computer Vision | GenAI Systems

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Ph.D. AI | Production ML | RAG | YOLO/OCR | Edge AI | Predictive ML | MLOps

anubhaparashar.github.io | GitHub | LinkedIn

AI engineer and research scientist with 10+ years across LLM/GenAI systems, computer vision, deep learning, NLP, edge AI and applied research. Builds LLM-agent workflows, RAG/offline retrieval systems, YOLO/OCR pipelines, predictive-maintenance platforms and cloud/edge AI deployments for industrial operations and research-to-production delivery.

PROFESSIONAL EXPERIENCE

Pearce Services, Gurugram

Jan 2025–Present

Industrial AI | LLM Agents | Computer Vision

Analytics & AI Engineer

- Built **production-oriented AI systems** for telecom field operations, infrastructure validation, predictive maintenance, routing optimization, edge vision and knowledge automation.
- Engineered **Execution-Aware Agentic VRP Studio** with **8 solver backends**, **18 disruption scenarios**, **40K orders / 400 routes**, locked-prefix re-optimization, Gemini agent layer and callable tools; achieved **100% assignment** with zero lateness/overtime in benchmark runs.
- Delivered **LLM/RAG automation** using **DeepSeek embeddings + FAISS** for offline knowledge retrieval; reduced manual content-update effort and improved retrieval accuracy for internal field workflows.
- Developed **YOLOv11 solar-defect detection** on **10K+ RGB/thermal images** with **mAP 0.92** and NVIDIA Jetson deployment; built **YOLO key-detection API** using FastAPI, Docker, Hugging Face and Azure Container Apps.
- Built **Pole Validation AI** using YOLO26n + GIS + OCR + multi-source reconciliation for ownership checks, geo-match, confidence scoring and Auto-Approve / Review / Reject decisions; created Generator Reliability dashboard with **14/30-day failure-risk scoring** in Streamlit/Snowflake.

Apr 2024–Dec 2024

LLM | OCR | Computer Vision

Research Scientist - AI & Metaverse

- Fine-tuned LLM workflow for vehicle RC queries, reducing manual search time by **70%**; Dockerized Gemini extraction APIs for document-intelligence workflows.
- Built **VITON-based virtual try-on** system with **95% user satisfaction**; created **PaddleOCR + Tesseract** extraction pipeline with **98% accuracy**.

Jul 2016–Apr 2024

AI | Deep Learning | Research

Manipal University Jaipur

Assistant Professor, Computer Science & Engineering

- Taught **AI, ML, Deep Learning, Computer Vision and NLP** for 8 years; supervised **15+ M.Tech theses** and mentored applied AI projects.
- Conducted Ph.D. research in **covariate-invariant gait biometrics** and surveillance AI; published **50+ papers**, contributed to **6 patents**, and served as IEEE/Springer PC member and reviewer.

TECHNICAL STRENGTHS

LLM Agents / GenAI: GPT-4o-mini, Gemini 2.0 Flash, DeepSeek, LangChain, tool use, function calling, DAG planning, MCP-style orchestration	RAG / Retrieval: FAISS, pgvector, embeddings, semantic metadata, offline retrieval, knowledge-base automation
Computer Vision: YOLOv11/YOLO26n, Ultralytics, OpenCV, OCR, PaddleOCR, Tesseract, VITON, RGB/thermal imagery, Jetson edge deployment	ML / MLOps: PyTorch, TensorFlow, Scikit-learn, XGBoost, GRU/LSTM, FastAPI, Streamlit, Docker, Azure, Hugging Face, Snowflake, CI/CD
Data / Systems: Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Redis, NetworkX, Plotly, REST APIs	Optimization: OR-Tools, PyVRP, OSRM, Esri ArcGIS, ant colony optimization, routing, scheduling, scenario benchmarking

RESEARCH, EDUCATION & CERTIFICATIONS

RESEARCH & PUBLICATIONS

50+ peer-reviewed papers across AI, computer vision, biometrics and intelligent systems; **6 granted/pending patents**; **600+ citations**; venues include **AI Review, Engineering Applications of AI, Neurocomputing** and **Pattern Recognition Letters**.

EDUCATION

Ph.D., CSE - Artificial Intelligence	
Manipal University Jaipur	2018–2023
M.Tech., CSE - Artificial Intelligence	
MD University, Rohtak	2014–2016
B.Tech., Computer Science & Engineering	
MD University, Rohtak	2009–2013

CERTIFICATIONS & ROLE MATCH

NVIDIA DeepStream Video Analytics on Jetson; AI on Jetson Nano; Deep Learning Specialization; Machine Learning Specialization; Data Science with Python & SQL.
Fit: LLM agents, RAG, multimodal/CV pipelines, evaluation-oriented ML systems and production delivery.

PROFILE KEYWORDS

LLM Agents Agentic AI RAG Multimodal AI Computer Vision YOLO OCR Edge AI Predictive Maintenance FastAPI Streamlit Docker Azure Snowflake Hugging Face MLOps